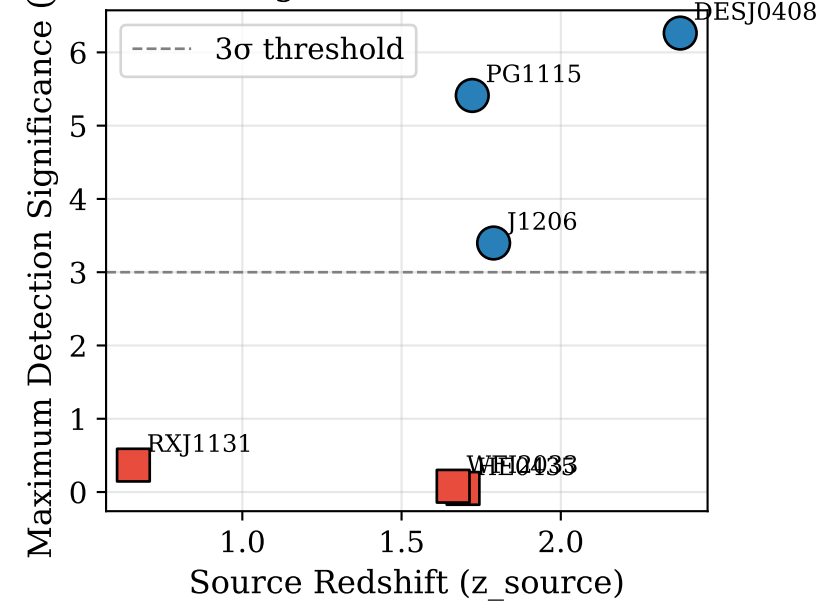
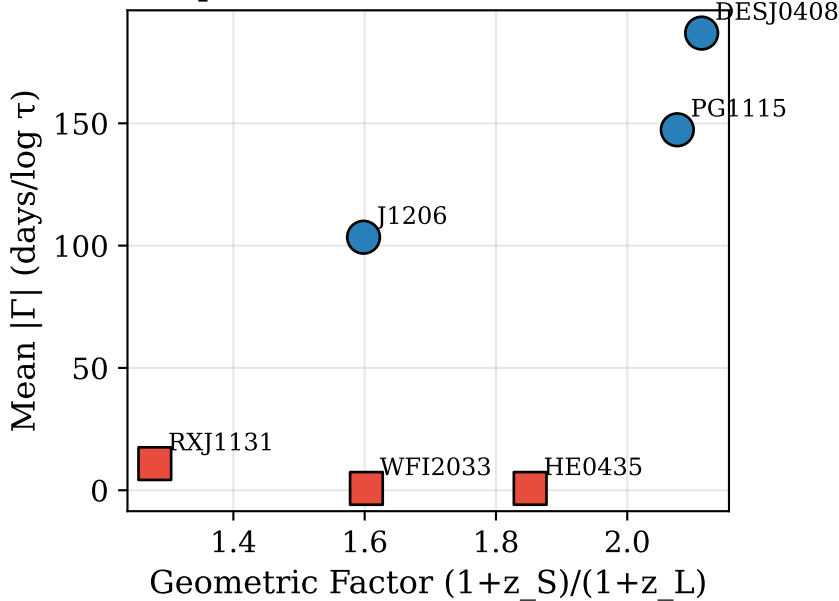


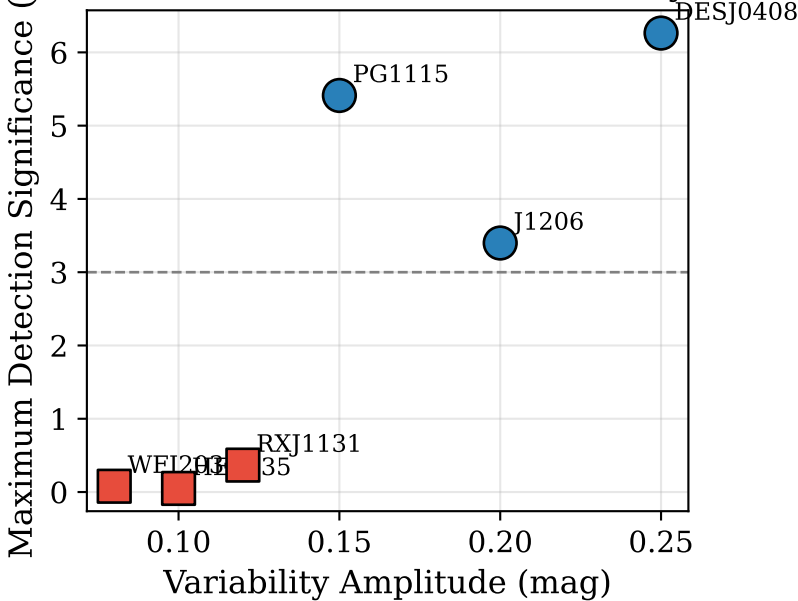
A) Detection Significance vs Source Redshift



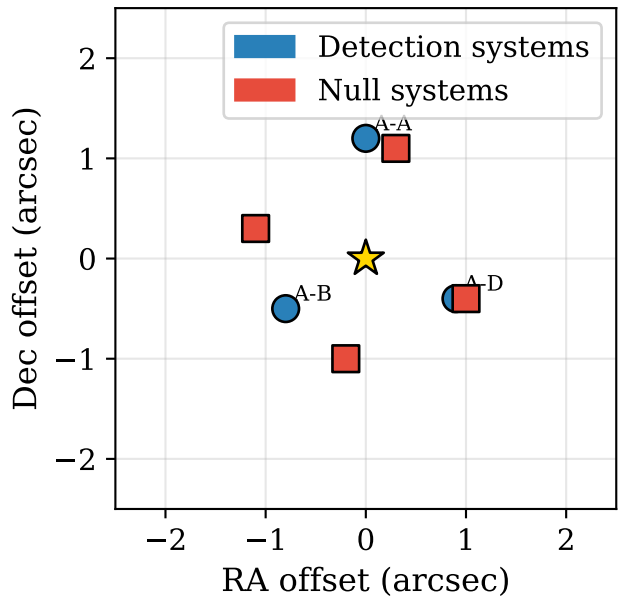
B. Temporal Shear vs Geometric Factor



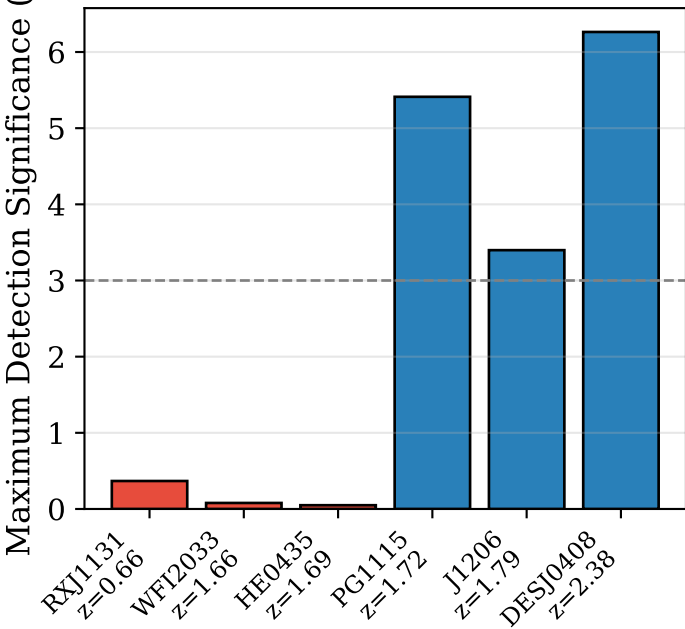
C. Detection vs Quasar Variability



D. Image Configurations



E. Systems Ordered by Source Redshift



F. Summary of Discriminants

KEY DISCRIMINANTS

DETECTION SYSTEMS (DESJ0408, PG1115, J1206):

- Higher source redshift ($z_S \sim 1.96$)
- Higher geometric factor $(1+z_S)/(1+z_L) \sim 2.0$
- Higher quasar variability amplitude

NULL SYSTEMS (HE0435, WFI2033, RXJ1131):

- Lower source redshift ($z_S \sim 1.34$)
- Lower geometric factor ~ 1.6
- Lower variability amplitude

SPECIAL CASE: RXJ1131

- Lowest z_{source} (0.658) in sample
- Shortest path through potential
- Expected to show weakest TEP signal

INTERPRETATION:

The TEP-GL temporal shear scales with PATH LENGTH through the gravitational potential:

$$\Gamma \propto (1+z_S)/(1+z_L) \times \int \nabla \Phi \cdot d\mathbf{l}$$

Higher $z_{\text{source}} \rightarrow$ longer path $\rightarrow$ larger Γ

This explains why:

- DESJ0408 ($z_S=2.38$) shows strongest signal
- RXJ1131 ($z_S=0.66$) shows null result